



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 65: Solar Playground with Security System Bi-Weekly Update 2

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Sponsor: Wonhyeok Jang
TA: Zian Wang

Project Summary

The SolarEye Security System is a solar-powered, autonomous surveillance solution for playgrounds and parks.

It enhances security with motion-activated cameras, real-time mobile alerts, LED deterrent lighting, and visitor charging stations.

This eco-friendly system ensures continuous, cost-effective protection for private parks and properties.



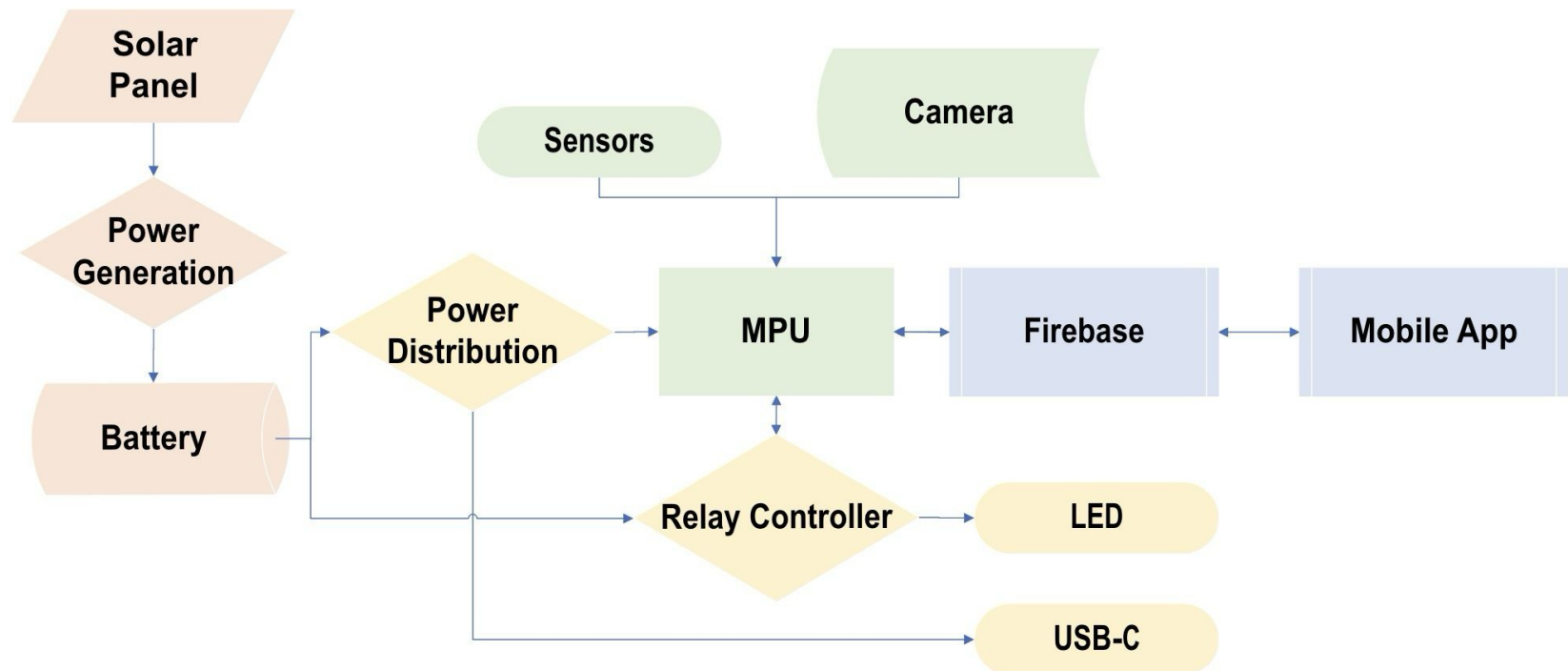
Project/Subsystem Overview

Red - Jackson Kitchel (Power Generation)

Yellow - Victor Igbokwe (Power Distribution)

Green - Alex Allahar (Microprocessor & Sensors)

Blue - Zayan Prasla (Network & Application)





Project Timeline

New PCBs Ordered/ Received (completed 02/03)	Integration of MPU and Application subsystems (to completed 02/25)	Integration of Power Delivery system with Microprocessor (to complete by 02/25)	Integration of Power Generation System with Power Delivery System (to complete by 02/28)	Final Integration (to complete by 03/03)	System Test (to complete by 03/10)	Validation (to complete by 03/20)	Demo and Report (to complete by 04/25)
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Power Generation

Jackson Kitchel

Accomplishments since last update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ New PCB and parts have been acquired ❖ Solar panel has been tested	❖ Finish soldering and testing ❖ Integrate with power distribution subsystem ❖ Integrate Voltage/current sensors with MCU

Power Generation

Jackson Kitchel

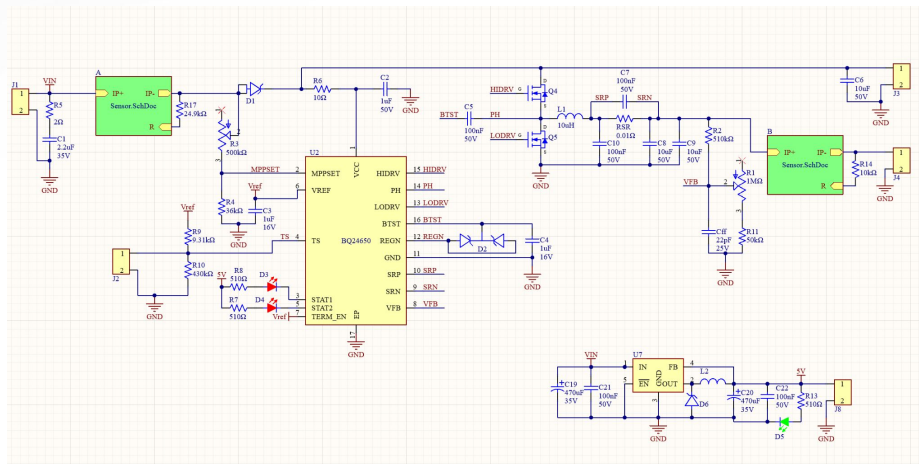


Figure 1: Charging Circuit Schematic

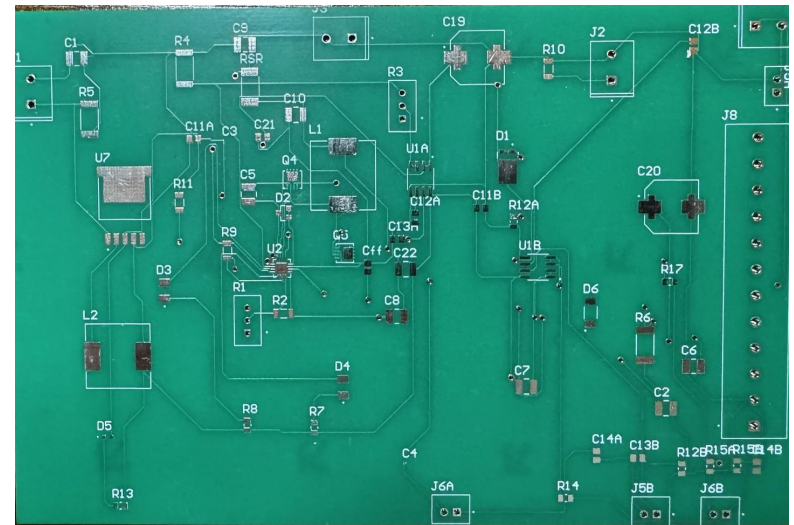


Figure 2: New PCB



Power Distribution

Victor Igbokwe

Accomplishments since last update 16 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Buck converter PCB soldered ❖ Tested Buck PCB with 12v input and various current values. ❖ Buck produced desired 5v output.	❖ Completed tests on the Buck converter in preparation for integrating with Microprocessor and Sensors subsystem. ❖ By next review relay control module PCB will be soldered and Tested. ❖ Begin Integration of Relay Controller PCB with Microprocessor and Sensors subsystem.

Power Distribution

Victor Igbokwe

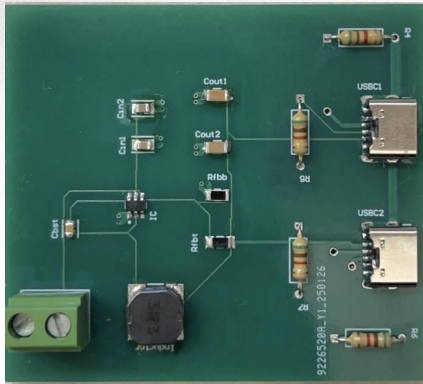


Figure 3: Buck PCB



Figure 4: Voltage Output from Buck converter

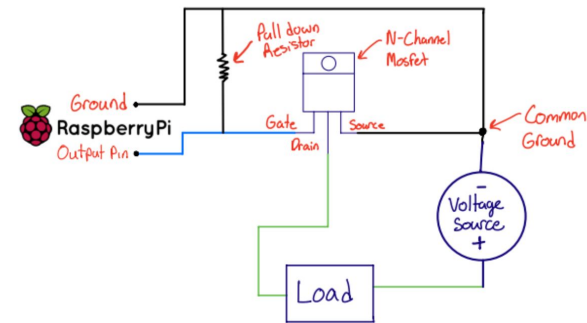


Figure 5: Schematic diagram Relay control module

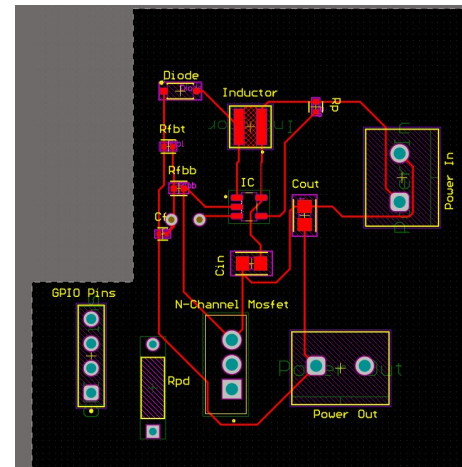


Figure 6: Altium PCB Design incorporating Relay control module



Microprocessor & Sensors

Alex Allahar

Accomplishments since last update 14 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Integrated Realtime Database to Firestore Database ❖ Validated PIR sensor communication to Firestore ❖ Developed the MOSFET Relay Code ❖ Update the UI of the Livestreaming	❖ Integrate Video Storage to App Database ❖ Validate Video Footage on Mobile App ❖ Integrate of Control module PCB with Microprocessor and Sensors subsystem. ❖ Control the LED via MPU



Microprocessor & Sensors

Alex Allahar

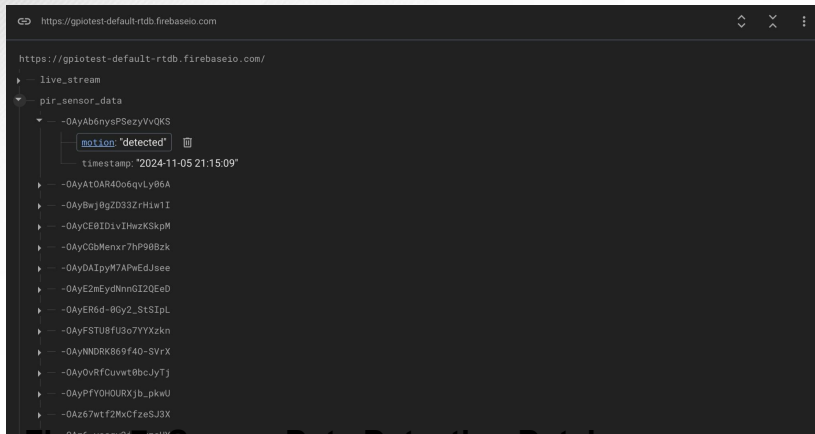


Figure 7: Sensor Data Detection Database

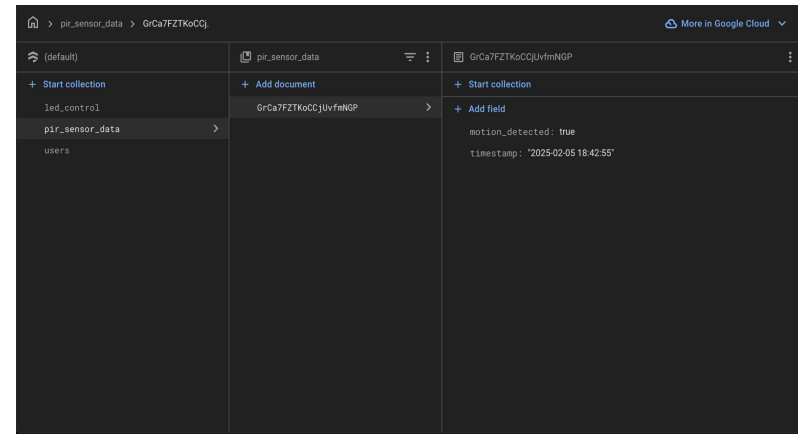


Figure 8: Sensor Data App Database

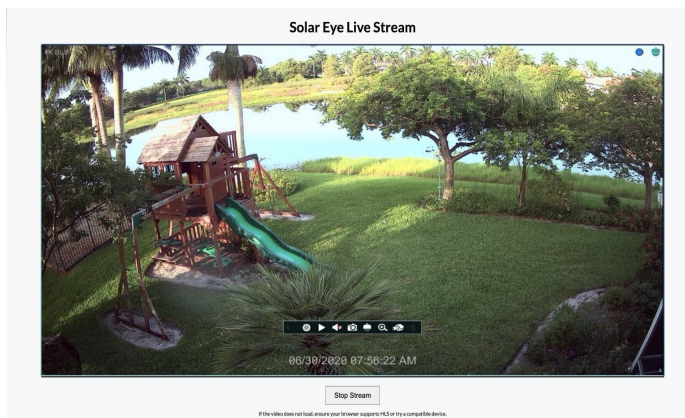


Figure 9: SolarEye Livestream Layout

```
1 import RPi.GPIO as GPIO
2 import time
3
4 RELAY_PIN = 17
5
6 GPIO.setmode(GPIO.BCM)
7 GPIO.setup(RELAY_PIN, GPIO.OUT)
8
9 def turn_on_led():
10     GPIO.output(RELAY_PIN, GPIO.HIGH)
11     print("LED is ON")
12
13 def turn_off_led():
14     GPIO.output(RELAY_PIN, GPIO.LOW)
15     print("LED is OFF")
16
17 try:
18     while True:
19         command = input("Enter 'on', 'off', or 'exit': ").strip().lower()
20         if command == "on":
21             turn_on_led()
22         elif command == "off":
23             turn_off_led()
24         elif command == "exit":
25             print("Exiting program...")
26             break
27         else:
28             print("Invalid command.")
29
30 except KeyboardInterrupt:
31     print("\nProgram interrupted.")
32
33 finally:
34     GPIO.cleanup()
35     print("GPIO cleaned up. Exiting...")
```

Figure 10: relay-control.py



Application and Network

Zayan Prasla

Accomplishments since last update 11 hrs of effort	Ongoing progress/problems and plans until the next presentation
❖ Refactored to use a date adapter instead of hard-coded dates for event detection. ❖ Fixed a security flaw where clicking password reset sent an email to all users. ❖ Ensured that sensitive information like passwords is not logged in clear text.	❖ Live Streaming ❖ UI update for the home page ❖ Testing MCU-App integration by uploading camera videos to Firebase and then to the app. ❖ Refactoring charging statistics to use real battery readings instead of hard-coded graphs. ❖ Connecting Camera/LED controls to Firebase to MCU



Application and Network

Zayan Prasla



Figure 11: Home Page

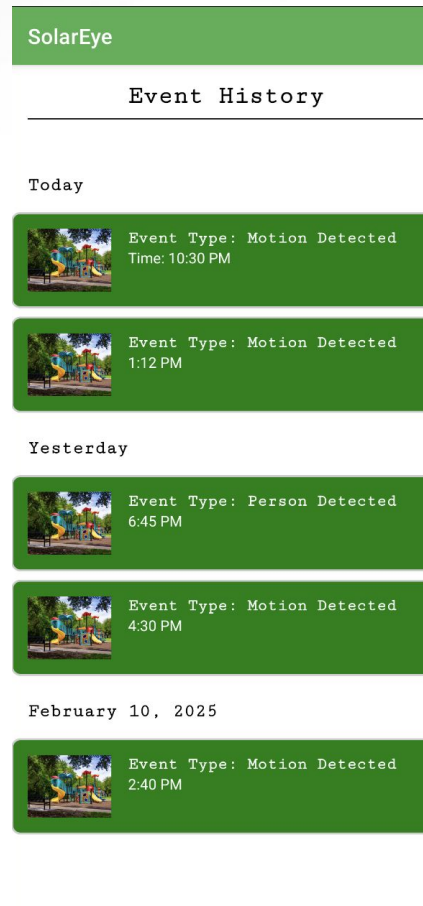


Figure 12: Event History

Hello,

Follow this link to reset your solareye-security-system password for your

https://solareye-security-system.firebaseio.com/_/auth/action?mode=apiKey=AlzaSyApdPKZizESib9-DxhlqVtvBDqzz_1RCiU&lang=en

If you didn't ask to reset your password, you can ignore this email.

Thanks,

Your solareye-security-system team

Figure 13: Password Reset Email

The Password Reset form is titled 'Reset your password' and is for the user 'zayanprasla@tamu.edu'. It includes a 'New password' input field with a toggle for visibility. Below the input field, there is a red error message: 'Missing password requirements: [Password must contain at least 6 characters, Password must contain an upper case character, Password must contain a numeric character, Password must contain a non-alphanumeric character]'. At the bottom right, there is a blue 'SAVE' button.

Figure 14: Password Reset

Execution & Plan

[illegible]



Texas Power and Energy Conference 02/10/2025

